

Leave Accrual Engine

Implementation Specification v1.3 | 2026-04-04
Parent Spec: f921f141-183f-4a23-bffe-9f52c739401a

Revision History

Ver	Date	Changes
1.0	04-03	Initial specification
1.1	04-04	Gap analysis: rate tiers, cycle constraints, mandatory reasons, sick leave clarification.
1.2	04-04	Cycle corrections: three cycle types, system-wide annual_leave_cycle_start, per-employee sick cycle, custom cycle anchors.
1.3	04-04	Per-employee rate override: optional annual_leave_override_days field. Tiers as default, override takes precedence.

1. Purpose

This document specifies how the leave accrual engine must behave. Every **[DECIDED]** decision is confirmed by the product owner and must not be re-interpreted.

2. Glossary

Term	Definition
Annual Leave Cycle	12-month period from system-wide annual_leave_cycle_start (default 1 Jan). Same for all employees. Also governs FRL.
Sick Leave Cycle	36-month period from each employee's employment_start_date. Per-employee, can start mid-month.
Custom Leave Cycle	Cycle with length and anchor defined per custom leave type by HR.
RATE	The monthly accrual rate for an employee: from per-employee override (if set), otherwise from tier lookup.
Active Days	Calendar days employed and not on accrual-pausing leave.
BCEA	Basic Conditions of Employment Act (South Africa).

3. Cycle Definitions

3.1 Annual Leave Cycle (System-Wide)

[DECIDED] System-wide setting, same for all employees. Default: 1 January. Must be 1st of a month.

Governs: **Annual Leave** and **FRL**. Cycle boundaries always align with month boundaries.

3.2 Sick Leave Cycle (Per-Employee)

[DECIDED] Always starts on exact employment_start_date, even mid-month. 36-month cycle.

3.3 Custom Leave Cycle (Per-Leave-Type)

[DECIDED] HR chooses cycle_anchor: calendar_year (system-wide) or employment_start_date (per-employee).

Cycle Summary

Leave Type	Cycle	Start Date	Length	Mid-Month ?
Annual Leave	System-wide	annual_leave_cycle_start	12 months	No
FRL	System-wide	Same as Annual	12 months	No
Sick Leave	Per-employee	employment_start_date	36 months	Yes
Custom (calendar)	System-wide	Same as Annual	Configurable	No
Custom (employment)	Per-employee	employment_start_date	Configurable	Yes
Maternity/Parental/ etc	No cycle	Event-triggered	Fixed	N/A
Unpaid	No cycle	Event-triggered	As approved	N/A

4. Accrual Run - Timing & Trigger

[DECIDED] Runs on the 1st of each month, credits the prior calendar month.

Idempotent via accrual_period (YYYY-MM). Includes terminated employees active during prior month. Sick 36-month resets checked for mid-month boundaries.

5. Annual Leave Accrual

5.1 Rate Determination (Two-Level System)

[DECIDED] Per-employee override takes precedence over tiers. When override is set, tiers are skipped entirely.

```
if employee.annual_leave_override_days is NOT NULL:
    RATE = annual_leave_override_days / 12
else:
    tier = highest where min_months <= completed months
    RATE = tier.annual_entitlement_days / 12
```

System-Wide Tiers (default)

Tier	Min Months	Annual Entitlement	Monthly Rate
1 (default)	0	15 days	1.25/month
2 (default)	24	20 days	1.6667/month

HR can add/edit/remove tiers. Mid-month tier crossing: entire month uses new rate.

Per-Employee Override

[DECIDED] Stored as annual_leave_override_days (nullable decimal, annual entitlement days). RATE = value / 12.

[DECIDED] Override is permanent until HR changes or clears it. Tier boundary crossings have no effect on overridden employees.

When NULL: tiers apply. When set (e.g., 18): RATE = 1.5, tiers ignored. Setting/clearing must be audit-logged with old value, new value, reason, and HR user.

The 'earned first' rule applies: employees may only take leave already accrued.

5.2-5.3 Full & Partial Month

```
Full: accrual = RATE
Partial: accrual = RATE * (active_days / calendar_days_in_month)
```

[DECIDED] Calendar-day pro-ration. No rounding - store exact decimals.

5.4 Termination

[DECIDED] Terminated employees receive final pro-rated accrual.

Must NOT skip terminated employees. Filter includes inactive with deactivation_date in prior month.

5.5-5.6 Unpaid & Accrual-Pausing Leave

Pauses: Unpaid, Maternity, Parental, Adoption, Commissioning, custom with pauses_annual_accrual=true.

Does NOT pause: Annual, Sick, FRL, custom unless configured.

Current implementation does not deduct pausing leave. Largest gap - must fix.

5.7-5.8 Combined Month & Settlement

$\text{active_days} = \text{days_employed} - \text{days_on_pausing_leave}$. On termination: credit immediately, mark final.

5.9 Rollover & Forfeiture

[DECIDED] Warnings at 60 and 30 days. Flag for HR at expiry - do NOT auto-forfeit.

6. Sick Leave Accrual

30 days per 36-month cycle (5-day worker). Anchored to exact employment_start_date.

6.2 First 6 Months

[DECIDED] Days worked = schedule minus recorded leave (not attendance).

```
days_worked = scheduled_working_days - leave_days_taken  
newly_earned = floor(cumulative / 26) - graduated_credited
```

Current impl uses 21.67 approximation. Must use actual leave records.

6.3 Transition

[DECIDED] On 1st of month after 6-month anniversary. Balance = full_entitlement - days TAKEN (not accrued).

6.4-6.5 Post-Transition & Reset

Single decrementing pool. At 36 months: reset. Graduated accrual does not re-apply. Mid-month boundary checked each run.

6.6 Medical Certificates

Statutory: >2 consecutive days. Company: Fri/Mon/adjacent to holiday.

7. Family Responsibility Leave

3 days/cycle (lump sum). Follows annual leave cycle. Requires 4+ months AND 4+ days/week.

[DECIDED] Resets automatically each cycle. Schedule drop mid-cycle: retain until cycle end.

Current impl: only checks months, never resets. Must fix both.

8. Maternity, Parental, Adoption & Commissioning

Type	Duration	Rules
Maternity	4 months	4 weeks before due. 6-week post-birth mandatory.
Parental	10 days	Non-birthing parent.
Adoption	10 weeks	One parent per event.
Commissioning	10 weeks	Surrogacy.

No accrual. Pauses annual leave. Overlapping leave blocked.

9. Custom Leave Types

Field	Type	Description
accrual_type	enum	none, lump_sum, monthly
accrual_rate	decimal	Days/month (monthly only)
lump_sum_amount	decimal	Days at cycle start
cycle_length_months	int	Cycle length
cycle_anchor	enum	calendar_year or employment_start_date
carry_over	bool	Carry to next cycle
pauses_annual_accrual	bool	Pauses annual accrual
requires_approval	bool	Requires approval
max_days_per_cycle	dec/null	Cap (null=unlimited)

pauses_annual_accrual and cycle_anchor missing from schema.

10. Unpaid Leave

No balance. Reduces annual accrual for the month (Section 5.5).

11. Manual Adjustments

[DECIDED] HR can adjust any balance. Reason mandatory. Effective immediately. Audit-logged.

12. Audit Trail

Field	Description
id	Unique identifier
timestamp	UTC timestamp
employee_id / leave_type	Affected employee and leave type
event_type	monthly_accrual, pro_rated_accrual, cycle_reset, forfeiture_warning, forfeiture_actioned, manual_adjustment, sick_leave_graduated_credit, sick_leave_transition, frl_cycle_reset, termination_settlement, sick_leave_cycle_reset, rate_override_changed
amount	Days credited/debited/forfeited
balance_before / after	Balance before and after
calculation_basis	Human-readable explanation
accrual_period	YYYY-MM (idempotency)
triggered_by	system or user_id
metadata	JSON context (e.g., rate_tier, override_days, old_override, new_override)

audit_logs table exists but is not written to. All paths must log.

13. Data Model

13.1 System-Wide Settings

Field	Type	Default	Description
annual_leave_cycle_start	month-day	Jan 1	Annual/FRL cycle start. Must be 1st of month.

13.2 Employee Profile

Field	Type	Description
employment_start_date	date	Immutable. Anchors sick leave cycle.
work_days_per_week	int (1-7)	Affects sick leave and FRL.
status	enum	active or inactive.
deactivation_date	date/null	For final settlement.
annual_leave_override_days	decimal/null	Optional override for annual entitlement (days/year). When set: RATE = value/12, tiers skipped. When null: tiers apply.

13.3-13.5 Balances, Sick Tracking, Rate Tiers

Key fields: current_cycle_balance, prior_cycle_carry_over (annual only), sick_cycle_start_date, graduated_accrual_active, cumulative_days_worked, graduated_days_credited. Rate tier table: id, min_months_of_service, annual_entitlement_days, monthly_accrual_rate.

Missing: separate balance fields, all sick tracking fields, rate tier table, annual_leave_override_days.

14. Accrual Run - Processing Order (12 Steps)

Step	Action	Details
1	Identify employees	All active during prior month, including terminated.
2	Determine RATE	If annual_leave_override_days set: RATE = value/12 (skip tiers). Otherwise: completed months -> highest matching tier.
3	Annual leave accrual	active_days minus pausing leave. RATE * (active/total). Idempotency. Audit.
4	Graduated sick leave	days_worked from schedule minus leave. Counter. Credit per 26. Audit.
5	Sick transitions	1st after 6-month anniversary. Balance = entitlement - TAKEN.
6	Sick 36-month resets	Check mid-month boundary. Reset to full entitlement.
7	Annual rollovers	Carry-over, zero current, schedule forfeiture warnings.
8	FRL resets	Reset to 3 (if eligible) or 0.
9	Custom monthly	Same pro-ration, per cycle_anchor.
10	Custom lump-sum	Grant at cycle boundary.
11	Forfeiture warnings	60 and 30 days before grace expiry.
12	Forfeiture flags	Flag for HR. Re-flag monthly.

15. Edge Cases (31)

#	Scenario	Expected
1	Starts last day of month	RATE * (1/31).
2	Starts 1st	Full RATE.
3	Full month unpaid	Accrual = 0.
4	Mid-month start + unpaid	Combined formula.
5	Terminated on 1st	1 day. Settlement.
6	Leap year Feb	Denom = 29.
7	4 months maternity	0 annual accrual x4.
8	Run twice	Idempotent.
9	No leave	Full accrual.
10	3 days/week	Sick=18d/36mo. FRL locked.
11	4-month mark	FRL unlocked. 3 days.
12	Sick grad + 5d leave	Counter reduced.

#	Scenario	Expected
1 3	Sick transition, 0 used	Balance = 30.
1 4	Accrued 4, used 1	29 (taken not accrued).
1 5	36mo reset, 10 unused	Resets. 10 lost.
1 6	Sick mid-month boundary	Reset on exact date.
1 7	HR skips forfeiture	Re-flagged monthly.
1 8	Reactivated	HR adjusts manually.
1 9	Custom pauses accrual	Days subtracted.
2 0	Multiple pausing types	Summed. No double-count.
2 1	work_days mid-month	End-of-month value.
2 2	Tier 1->2 mid-month (no override)	Entire month Tier 2.
2 3	Non-1st cycle start	Rejected.
2 4	Blank adjustment reason	Rejected.
2 5	Terminated before month	Skipped.
2 6	Custom employment-anchored	Pro-rated or exact date.
2 7	HR changes cycle start	Bridged. Confirmation.
2 8	Override = 18 days	RATE = 1.5. Tiers skipped.
2 9	Override + tier boundary	Override prevails. No change.
3 0	Override cleared	Falls back to tiers.
3 1	Override changed pre-accrual	Latest value used. Both logged.

16. Notifications

Trigger	To	Content
Monthly accrual	Employee	[X] days accrued. Balance: [Y].
Forfeiture 60d	Emp, HR	[X] days subject to forfeiture on [date].
Forfeiture 30d	Emp, HR	Same, urgent.
Forfeiture flag	HR	Past grace. Action required.
Sick transition	Emp, HR	Entitlement updated to [X] days.
Sick reset	Emp, HR	Cycle reset. [X] days.
FRL reset	Employee	Reset to 3 days.
FRL unlocked	Employee	Now eligible.
Termination	HR	Final accrual processed.
Rate change	Emp, HR	Rate changed to [X] days/year.

17. Testing Requirements (32 Tests)

#	Test	Expected
1	Tier 1 full month	1.25
2	Tier 2 full month	1.6667
3	Starts 15th/30d	$RATE * (16/30)$
4	Terminated 10th/31d	$RATE * (10/31)$
5	7d unpaid/28d	$RATE * (21/28)$
6	Full month maternity	0
7	Sick grad: 22d, no leave	+22
8	Sick grad: 5d leave	-5
9	Sick transition	entitlement-TAKEN
10	Sick 36mo reset	Full
11	Sick mid-month reset	Correct
12	FRL: 3mo29d	Locked
13	FRL: 3d/wk, 6mo	Locked
14	FRL reset eligible	3
15	FRL reset 3d/wk	0
16	Annual rollover	Zeroed+carry
17	Forfeiture flag	Flag, not zero

#	Test	Expected
18	Manual +/- reason	Audit
19	Manual blank reason	Rejected
20	Idempotency	No dupes
21	Leap Feb	Denom=29
22	Starter+unpaid	Both applied
23	Custom monthly	Pro-rated
24	Custom employment	Per-employee
25	23mo vs 24mo (no ovr)	Tier 1->2
26	Cycle start validation	Reject non-1st
27	Terminated final month	Included
28	Terminated before	Skipped
29	Override=18	RATE=1.5, skip tiers
30	Override+tier boundary	Override prevails
31	Override cleared	Tiers resume
32	Override change logged	Audit with old/new

Appendix A: Gap Cross-Reference

Gap	Description	Spec	Status
1	Accrual timing	S4	Change to 1st.
2	Pausing leave	S5.5-5.6	Must deduct.
3	Idempotency	S4,S12	accrual_period.
4	Audit trail	S12	All paths.
5	Termination	S5.4,5.8	Don't skip.
6	Rollover	S5.9	Zero+warnings.
7	FRL eligibility	S7.2	Check work_days.
8	FRL reset	S7.3	At boundary.
9	Rate 24mo	S5.1	Tiers+override (v1.3).
10	Schema	S13	All fields.
11	Sick graduated	S6.2	Actual records.
12	Sick transition	S6.3	Days TAKEN.
13	Mid-month split	S3.1,6.5	Eliminated/handled.

Gap	Description	Spec	Status
14	pauses field	S9.1	Add to schema.

Appendix B: Cycle Summary

Leave Type	Cycle	Start	Length	Mid-Month ?
Annual	System-wide	annual_leave_cycle_start	12mo	No
FRL	System-wide	Same as Annual	12mo	No
Sick	Per-employee	employment_start_date	36mo	Yes
Custom (cal)	System-wide	Same as Annual	Config	No
Custom (emp)	Per-employee	employment_start_date	Config	Yes
Maternity etc	No cycle	Event	Fixed	N/A
Unpaid	No cycle	Event	Approved	N/A